

Race in time following Erdős

1946: $c\sqrt{n}$

1952 Leo Moser

———— $cn^{2/3}$

1984 Fan Chung

———— $cn^{5/7}$

1992 Endre Szemerédi

———— $cn^{4/5}/\log n$

1993 László Székely

———— $cn^{4/5}$

2001 József Solymosi

———— $cn^{6/7}$

2003 Gábor Tardos

———— $cn^{4e/(5e-1)-\varepsilon}$

2004 Nets Katz

———— $cn^{(48-14e)/(55-16e)-\varepsilon}$

2015 Larry Guth

———— $cn/\log n$

Future... ?